

DOCUMENT 07 · INPUT AND OUTPUT CONTRACTS

# Input and output contracts

The mandatory structure of every SPAD artefact, readable by people and by machines, so that the &quot;incomplete artefact&quot; validation can be automated and agents can chain phases without losing information

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**Version under review: please do not circulate.** The current state of SPAD (version 0.x) is not meant to be shared widely. It is public so that a small number of people can review it, give feedback and help improve it. Documents and tools are being adapted to make them reusable; this notice will disappear when the framework reaches version 1.x.

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## 01

# Purpose

Each SPAD phase produces an artefact with a **contractual structure**: mandatory sections and fields that the instruction requires (document 06) and whose absence invalidates the response (document 03). This document defines that structure in JSON, so that:

- a person can check at a glance that the output is complete;
- an automatic validator can reject an incomplete artefact before anyone sees it;
- an agent that chains phases receives from the previous one exactly what it needs.

The artefact may be presented in Markdown with the same sections; JSON is the canonical form for validation and exchange.

**Why it matters.** "Incomplete artefact" is the most frequent cause of invalidation and the easiest to automate. An explicit contract turns that check into a mechanical step and leaves to the person the judgement that does require a person: whether the content is correct.

## 02

# Common conventions

## 2.1 Common block

Every artefact includes these fields:

```
{
  "spad": { "contract_version": "0.1" },
  "topic": "string, snake_case, maximum 20 characters",
  "phase": "PLAN | AUDIT_PLAN | CODE_PRIMER | TEST_STRATEGY | IMPLEMENTATION | TEST_IMPLEMENTATION | ",
  "iteration": "integer from 1",
  "date": "ISO 8601",
  "model": {
    "provider": "string",
    "model": "string",
    "model_version": "string",
    "role": "planner | reviewer | builder | fixer | analyst | diagnostic | security_reviewer",
    "instruction": "identifier and version of the instruction used",
    "global_context": "version",
    "project_context": "version"
  },
  "deviations": [
    { "situation": "string", "decision_requested": "string" }
  ]
}
```

**deviations** records the situations that the phase could not resolve and on which a decision is requested; it never contains decisions taken by the AI.

## 2.2 Scales

| Scale                   | Values                                                                                                  |
|-------------------------|---------------------------------------------------------------------------------------------------------|
| Verdict                 | GO · GO_WITH_CHANGES · NO_GO (all reviews, including the security review and the behaviour evaluation). |
| Severity                | CRITICAL · HIGH · MEDIUM · LOW .                                                                        |
| Probability             | HIGH · MEDIUM · LOW .                                                                                   |
| Recommendation priority | MUST · SHOULD · NICE_TO_HAVE .                                                                          |
| Quality                 | EXCELLENT · GOOD · ADEQUATE · POOR · MISSING .                                                          |

## 2.3 Human validation record

It is not part of the AI's artefact: the person writes it in the topic's `VALIDATION.md` (document 02, section 4.2). Its canonical form:

```
{
  "artefact": "SPAD_02_AUDIT_PLAN",
  "person": "identifier",
  "date": "ISO 8601",
  "outcome": "VALID | INVALID | EXCEPTION_APPROVED",
  "invalidation_cause": "PHASE_VIOLATION | INCOMPLETE_ARTEFACT | OFF_PLAN | NO_PRIOR_REVIEW | SELF_API",
  "ai_verdict": "GO | GO_WITH_CHANGES | NO_GO | null",
  "verdict_accepted": "true | false | null",
  "observations": "string"
}
```

### 03

## Main cycle contracts

### 3.1 PLAN

```
{
  "architecture": { "overview": "string", "patterns": ["string"] },
  "components": [
```

```

    { "name": "string", "responsibility": "string", "dependencies": ["string"], "interfaces": ["string"] },
    "flows": [
      { "name": "string", "trigger": "string", "steps": ["string"], "data": "string" }
    ],
    "decisions": [
      { "decision": "string", "justification": "string", "alternatives": ["string"], "trade_offs": "string" }
    ],
    "risks": [
      { "risk": "string", "severity": "CRITICAL|HIGH|MEDIUM|LOW", "probability": "HIGH|MEDIUM|LOW", "mitigation": "string" }
    ],
    "ai_system": {
      "applies": "true | false",
      "autonomy_level": "A0 | A1 | A2 | A3",
      "ai_components": ["model, instructions, retrieval, tools, memory"],
      "action_limits": ["string"],
      "data_to_model": { "provider": "string", "information_classes": ["string"] },
      "failure_modes": [ { "failure": "string", "safe_behaviour": "string" } ],
      "model_change": "strategy for provider updates"
    }
  }
}

```

## 3.2 AUDIT\_PLAN

```

{
  "issues": [
    { "id": "string", "category": "COUPLING|COHESION|SCALABILITY|CONCURRENCY|SECURITY|OBSERVABILITY|OTHER",
      "severity": "CRITICAL|HIGH|MEDIUM|LOW", "description": "string", "location": "section of the plan" },
    ...
  ],
  "recommendations": [
    { "issue": "id", "recommendation": "string", "priority": "MUST|SHOULD|NICE_TO_HAVE" }
  ],
  "exceptions_evaluated": [
    { "global_rule": "string", "exception": "string", "evaluation": "string" }
  ],
  "verdict": "GO|GO_WITH_CHANGES|NO_GO",
  "justification": "string",
  "next_steps": ["string"]
}

```

Consistency rule: `verdict = GO` is incompatible with any issue of severity `CRITICAL` or `HIGH`.

### 3.3 CODE\_PRIMER

```
{
  "structure": { "directories": "tree", "modules": "string" },
  "rules": [
    { "category": "NAMING|STRUCTURE|ERROR_HANDLING|CONCURRENCY|TESTING|DOCUMENTATION|LOGGING|OTHER",
      "rule": "string", "examples": ["string"], "enforcement": "MANDATORY|RECOMMENDED" }
  ],
  "contracts": [
    { "type": "API|INTERFACE|DATA_STRUCTURE|EVENT", "definition": "string", "validation": ["string"] }
  ],
  "anti_patterns": [
    { "pattern": "string", "why": "string", "alternative": "string" }
  ],
  "ai_rules": {
    "system_instructions": ["string"], "tool_contracts": ["string"], "interaction_logging": ["string"]
  }
}
```

### 3.4 TEST\_STRATEGY

```
{
  "coverage": { "minimum_total": "percentage", "minimum_unit": "percentage", "minimum_integration": "percentage" },
  "layers": [ { "layer": "UNIT|INTEGRATION|E2E|PERFORMANCE|SECURITY", "purpose": "string", "tools": ["string"] } ],
  "cases": [
    { "id": "string", "name": "string", "type": "UNIT|INTEGRATION|E2E", "component": "string",
      "scenario": "string", "expected_outcome": "string", "priority": "CRITICAL|HIGH|MEDIUM|LOW" }
  ],
  "doubles_and_data": [ { "name": "string", "purpose": "string", "structure": "string" } ],
  "edge_cases": [ { "case": "string", "approach": "string" } ],
  "behaviour_evaluation": {
    "applies": "true | false",
    "sets": ["identifier and version"],
    "criteria": [ { "metric": "string", "threshold": "value" } ],
    "adversarial": ["string"],
    "bias": ["string"]
  }
}
```

### 3.5 IMPLEMENTATION

```
{
  "files_created": [ { "path": "string", "purpose": "string", "key_functions": ["string"] } ],
}
```

```

"files_modified": [ { "path": "string", "changes": "string", "reason": "string" } ],
"dependencies_added": [ { "name": "string", "version": "string", "licence": "string", "reason": "string" } ],
"compliance": { "follows_primer": "true | false", "follows_plan": "true | false" },
"next_step": "TEST_IMPLEMENTATION"
}

```

Every deviation goes in the common block `deviations`, unresolved.

### 3.6 TEST\_IMPLEMENTATION

```

{
  "tests": [ { "file": "string", "cases": ["id"], "type": "UNIT|INTEGRATION|E2E" } ],
  "doubles_and_data": [ { "file": "string", "purpose": "string" } ],
  "execution": { "command": "string", "variables": ["string"], "requirements": ["string"] },
  "coverage_obtained": { "total": "percentage", "unit": "percentage", "integration": "percentage" },
  "compliance": { "follows_strategy": "true | false" }
}

```

### 3.7 AUDIT\_TESTS

```

{
  "coverage": { "actual_total": "percentage", "expected_total": "percentage", "actual_unit": "percentage" },
  "quality": [ { "aspect": "ASSERTIONS|EDGE_CASES|ERROR_PATHS|DOUBLES|DATA|CLARITY", "status": "EXCELLENT|GOOD|ADEQUATE" },
  "case_review": [ { "id": "string", "implemented": "true | false", "quality": "EXCELLENT|GOOD|ADEQUATE" },
  "verdict": "GO|GO_WITH_CHANGES|NO_GO",
  "justification": "string",
  "required_fixes": [ { "issue": "string", "severity": "CRITICAL|HIGH|MEDIUM|LOW", "recommendation": "string" } ]
}

```

### 3.8 AUDIT\_CODE

```

{
  "fidelity_to_plan": { "score": "0-100", "deviations": [ { "deviation": "string", "justified": "true | false" } ],
  "primer_compliance": { "score": "0-100", "violations": [ { "rule": "string", "location": "string", "severity": "CRITICAL|HIGH|MEDIUM|LOW" } ],
  "technical_risks": [ { "risk": "string", "category": "PERFORMANCE|SECURITY|RELIABILITY|MAINTAINABILITY", "severity": "CRITICAL|HIGH|MEDIUM|LOW" } ],
  "technical_debt": [ { "item": "string", "impact": "string", "recommendation": "string" } ],
  "dependencies": [ { "name": "string", "exists": "true | false", "provenance": "string", "vulnerability": "string" } ],
  "findings": [ { "finding": "string", "severity": "CRITICAL|HIGH|MEDIUM|LOW", "location": "string", "recommendation": "string" } ],
  "verdict": "GO|GO_WITH_CHANGES|NO_GO",
  "justification": "string",
}

```

```
"next_step": "VERSION|FIX_PRIMERS|PLAN"
}
```

### 3.9 FIX\_PRIMERS

```
{
  "fixes": [
    { "id": "string", "finding": "id of the finding", "problem": "string", "file": "string", "location": "string", "change": "string", "justification": "string", "impact": "string", "regression_risk": "LOW|MEDIUM|HIGH" },
  ],
  "files_modified": ["string"],
  "tests_updated": ["string"],
  "overall_regression_risk": "LOW|MEDIUM|HIGH",
  "next_step": "AUDIT_CODE"
}
```

### 3.10 VERSION

```
{
  "version": { "number": "MAJOR.MINOR.PATCH", "type": "MAJOR|MINOR|PATCH", "justification": "string" },
  "changes": [ { "category": "ADDED|CHANGED|DEPRECATED|REMOVED|FIXED|SECURITY", "description": "string" } ],
  "breaking_changes": [ { "change": "string", "impact": "string", "migration": "string" } ],
  "migration": { "required": "true | false", "steps": ["string"], "effort": "string" },
  "deployment": { "prerequisites": ["string"], "plan": "string", "rollback": "string", "rollback_test": "string" },
  "ai_system": { "model": "string", "parameters": "string", "instructions_version": "string", "evaluation": "string" }
}
```

## 04

# Complementary cycle contracts

### 4.1 DOCUMENT\_EXISTING (L1)

```
{
  "code_location": "string",
  "behaviour": { "purpose": "string", "functionality": ["string"], "inputs": ["string"], "outputs": ["string"] },
  "components": [ { "name": "string", "type": "CLASS|FUNCTION|MODULE|SERVICE", "responsibility": "string" } ],
  "dependencies": { "internal": ["string"], "external": ["string"] },
  "diagrams": [ { "type": "FLOW|ARCHITECTURE|SEQUENCE", "file": "string", "description": "string" } ]
}
```

```
"unverified": ["statements inferred without execution evidence"],
"risks": ["string"]
}
```

## 4.2 DISCOVERY\_LEGACY (L2)

```
{
  "change_objective": "string",
  "areas_affected": [ { "component": "string", "impact": "CRITICAL|HIGH|MEDIUM|LOW", "reason": "string" },
  "modification_risks": [ { "risk": "string", "severity": "CRITICAL|HIGH|MEDIUM|LOW", "mitigation": "string" },
  "tests": { "existing": ["string"], "coverage": "percentage", "gaps": ["string"], "characterisation": "string" },
  "edge_cases": [ { "case": "string", "current_behaviour": "string", "has_test": "true | false" } ],
  "recommendations": ["string"],
  "next_step": "PLAN"
}
```

## 4.3 DEBUG (D1)

```
{
  "incident": { "date": "ISO 8601", "description": "string", "environment": "DEVELOPMENT|STAGING|PRODUCTION",
  "mode": "STATIC|WITH_EXECUTION",
  "evidence": { "files": ["string"], "commands": ["string"], "logs": ["string"], "traces": ["string"] },
  "root_cause": { "summary": "string", "explanation": "string", "category": "LOGIC|CONFIGURATION|CONCURRENCY",
  "verdict": "CODE|CONFIGURATION|INFRASTRUCTURE",
  "actions": [ { "action": "string", "priority": "CRITICAL|HIGH|MEDIUM|LOW", "next_step": "PLAN|HOTFIX" },
}
```

## 4.4 HOTFIX (U2–U7)

```
{
  "priority": "P1|P2",
  "fix_type": "TEMPORARY|DEFINITIVE",
  "root_cause": "summary of the D1 report",
  "simplified_plan": { "change": "string", "scope": "string", "risk": "string", "minimal_tests": ["string"] },
  "accelerated_review": { "verdict": "GO|GO_WITH_CHANGES|NO_GO", "justification": "string" },
  "implementation": { "files": ["string"], "changes": "string", "tests_executed": ["string"] },
  "deployment": { "date": "ISO 8601", "method": "string", "emergency_approval": "person", "rollback_available": "boolean",
  "monitoring": { "duration_minutes": "integer", "indicators": ["string"], "status": "STABLE|UNSTABLE|DEGRADED",
  "post_mortem": { "timeline": ["string"], "root_cause": "string", "effectiveness": "SOLVED|MITIGATED|PENDING" },
}
```

```
"definitive_plan": { "required": "true | false", "topic": "string" }
}
```

## 4.5 SECURITY\_AUDIT (S1)

```
{
  "scope": { "files": ["string"], "components": ["string"] },
  "vulnerabilities": [
    { "id": "string", "severity": "CRITICAL|HIGH|MEDIUM|LOW",
      "category": "INJECTION|XSS|AUTHENTICATION|AUTHORISATION|CRYPTOGRAPHY|SECRETS|INPUT_VALIDATION|A",
      "reference": "entry in the reference list", "description": "string", "location": "string", "exp:
    ],
  "secrets": { "found": "true | false", "locations": ["string"], "recommendation": "string" },
  "vulnerable_dependencies": [ { "name": "string", "version": "string", "vulnerability": "identifier" },
  "deployment_decision": { "critical": "integer", "high": "integer", "medium": "integer", "low": "integer" },
  "proposed_residual_risks": [ { "risk": "string", "severity": "MEDIUM|LOW", "recommendation": "string" },
  "verdict": "GO|GO_WITH_CHANGES|NO_GO"
}
```

Acceptance of the residual risks is not part of the contract: a person writes it in the validation record.

## 4.6 BEHAVIOUR\_EVAL (E)

```
{
  "set": { "id": "string", "version": "string", "cases": "integer" },
  "results": [ { "metric": "string", "value": "number", "threshold": "number", "meets": "true | false" },
  "failures": [ { "category": "string", "count": "integer", "examples": ["string"] } ],
  "adversarial": [ { "test": "string", "outcome": "PASSED|FAILED", "detail": "string" } ],
  "bias": [ { "test": "string", "segments": ["string"], "difference": "number", "threshold": "number" },
  "human_review_sample": { "cases": "integer", "selection_criterion": "string" },
  "verdict": "GO|GO_WITH_CHANGES|NO_GO",
  "justification": "string",
  "return_phase": "PLAN|IMPLEMENTATION|null"
}
```

## 05

# Format of the contexts

The contexts are not phase artefacts and are written in Markdown with fixed sections (document 02):

```
# GLOBAL_CONTEXT (version n)
## Artefacts
## Tests
## Versioning
## Baseline security
## Data and model providers
## Prohibitions
## Parameters of the complementary cycles
## Approved models

# PROJECT_CONTEXT (version n)
## Technology
## Architecture
## Domain rules
## Compliance
## Team conventions
## Deployment
## Justified exceptions
```

## 06

# Automatic validation

An artefact validator checks, before the human validation:

| Check                                                                                                                 | Outcome if it fails  |
|-----------------------------------------------------------------------------------------------------------------------|----------------------|
| Common block present and complete, including <code>model</code> .                                                     | Incomplete artefact. |
| All the mandatory sections of the phase's contract.                                                                   | Incomplete artefact. |
| Verdict, severity and priority values within the scales.                                                              | Incomplete artefact. |
| <code>verdict = GO</code> with open critical or high issues.                                                          | Self-approval.       |
| <code>topic</code> in the correct format and matching the folder.                                                     | Incomplete artefact. |
| <code>iteration</code> consecutive with the previous artefact of the same phase.                                      | Incomplete artefact. |
| Cross-references ( <code>finding</code> , <code>case</code> , <code>issue</code> ) existing in the previous artefact. | Incomplete artefact. |

The validator **does not judge the content**: that is for the AI Reviewer and the person. A serverless tool with this function is part of the next steps of SPAD (document 08).

# Related documents

| Document                                             | Relationship                                                            |
|------------------------------------------------------|-------------------------------------------------------------------------|
| document 01 · Operating guide                        | Phases that produce each artefact.                                      |
| document 02 · Contexts, topics and artefact register | Model record and validation record.                                     |
| document 03 · Validation policy                      | Causes that the automatic validator detects.                            |
| document 04 · Complementary cycles                   | Contracts L1, L2, D1, U and S1.                                         |
| document 05 · Systems that include AI                | Contract of the behaviour evaluation and <code>ai_system</code> fields. |
| document 06 · Instructions by phase                  | Instructions that require these outputs.                                |

# Version control

| Version | Date       | Changes                                                                                                                                                                                                                                                                                                                  |
|---------|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0.1     | 19-09-2026 | First version. Rewrites the previous specification: common block with model record and deviations; single scales; human validation record; main cycle contracts with dependencies and licences; complementary cycle contracts; behaviour evaluation contract; format of the contexts; checks of the automatic validator. |